

UNITED INTERNATIONAL UNIVERSITY

Course Outline: BST 2216/QNT 3307 (Business Statistics II/Applied Business Statistics)
Credits: 03

In a competitive and ever-changing business landscape, good decision making relies on timely data, its analysis and insights gained from it. This course helps students develop their understanding how informed decisions are being made from data insights, and to communicate the results to its stakeholders for effective decision making. The main focus of this course will be on inferential statistical techniques and their application in business decisions. Rather than pure theories and mathematical calculations, the course will follow a hands-on approach to understand and analyze data in order to interpret and gain insights for various stakeholders, as well as orient students with latest software and tools including SPSS, Python for data science applications, and machine learning applications in analyzing data for business insights. (Pre-requisite: Business Statistics I, BST 1308).

Course Objectives:

1. Understand basic concepts of inferential statistics
2. Understand the process and application of hypothesis testing
3. Understand the use of statistical tools and techniques for business research
4. Apply statistics to solve business problems
5. Gain basic software and programming skills for expertise in data science and analytics

Tentative Contents:

- Introduction and basic review of statistics, data types and descriptive statistics
- Basics of inferential statistics, statistical distributions
- one-sample test of hypothesis
- Further topics on hypothesis testing, two-sample test of hypothesis
- Analysis of Variance (ANOVA)
- Linear Regression and Correlation analysis
- Non-parametric methods and applications
- Further topics in regression analysis

Recommended Texts:

- Statistics for Management (8th Edition), Richard I. Levin, David S. Rubin, Masood H. Siddiqui, Sanjay Rastogi, Pearson Education.

Recommended Software: SPSS (version 25 or higher), Python 3, Jupyter Notebook.

Google Colab will be used for demonstration purposes, and no prior knowledge of programming or any statistical software is assumed. No worries!!

Important: Various course materials will be provided in the form of samples, data sets, programming codes and guidelines for practical analysis and skill enhancement.

General Rules to be followed:

- Please be on time for your classes. If you are late for any reason, please enter quietly without asking for permission. Do not interrupt the session in progress. During the class, if needed, you may also go out quietly and come back quietly without interrupting the session.
- Attendance is automatically recorded through scanning, therefore, if there is any attendance issue then students must inform the instructor instantly during or immediately after the class, otherwise it cannot be recorded or amended later. Please check and ensure your attendance record everyday.
- Please make all of your electronic devices silent and do not use them during the class. You may leave the classroom in emergency cases, for the purpose of using mobile phones.
- You are most welcome to ask question/put comments during the class session by raising your hand first. Class participation is encouraged and rewarded.
- There will be no make-up of quizzes, no matter what the reason could be. Mid-term examination, if missed only on valid and documented reasons, may be retaken upon departmental approval and on a different syllabus. Final examination, if missed only on valid and documented reasons (as may be approved by the department and on a different syllabus), may be retaken on a case-by-case basis.
- Students are expected to abide by the university rules and regulations, cherish common human values, promote ethics and enhance the image of UIU both in and off campus.

Class Schedule : Sat/Tue (9:51AM-11:10AM), Lab 524
Counseling hours : Sat/Tue (11:11AM-12:30PM), Room 333(C)

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Course materials can be found at the ELMS site of UIU. All students are strongly encouraged to use ELMS to stay updated on the course.

Professional Blog: <http://khandoker.wordpress.com>

(Note: Whenever you contact through SMS, Email or Whatsapp group, please include your name, ID and section when you contact)

Best Wishes!